



CUSTOMER CHURN PREDICTION & REVENUE PROTECTION STRATEGY

Python | CatBoost | SHAP | Power BI



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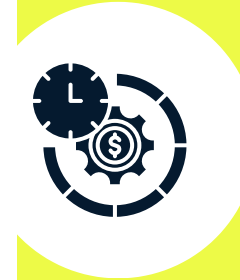
EXECUTIVE SUMMARY

PROJECT BACKGROUND



- Telco company's 26.54% churn rate leading to \$27,681 in monthly revenue at risk.
- Goal: Transition from reactive service to a data-driven Precision Retention Strategy.

CHURN RETENTION STRATEGIES



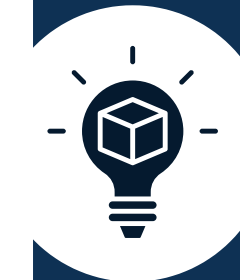
- Transitioned from a "one-size-fits-all" approach to a tiered, data-driven framework
- Precision Strategy (Optimized):
Focus: ROI & Revenue Protection.
Method: Targets "Whales" (High-Value/High-Risk) using specific service gap insights

ANALYSIS & EVALUATION



- Deployed CatBoost for superior handling of categorical data.
- Performance: Achieved 81% Accuracy and 82% Recall to identify churn before it happens.
- SHAP analysis isolated Monthly Charges (Tenure, and Contract Type as the primary triggers.

IMPACT SIMULATION



- Precision Strategy: 20 customers targeted
- Net Profit: +\$200.40
- ROI: +99.88%

BUSINESS PROBLEMS

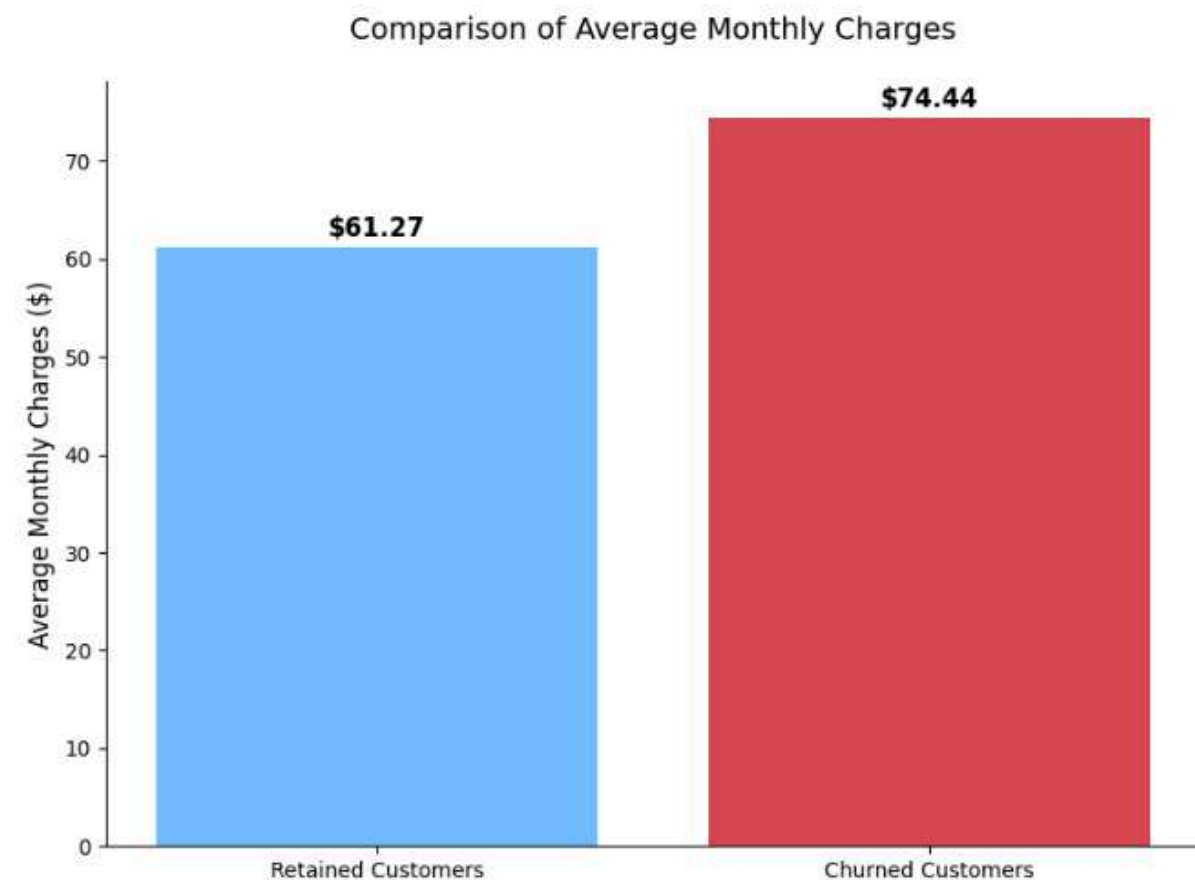
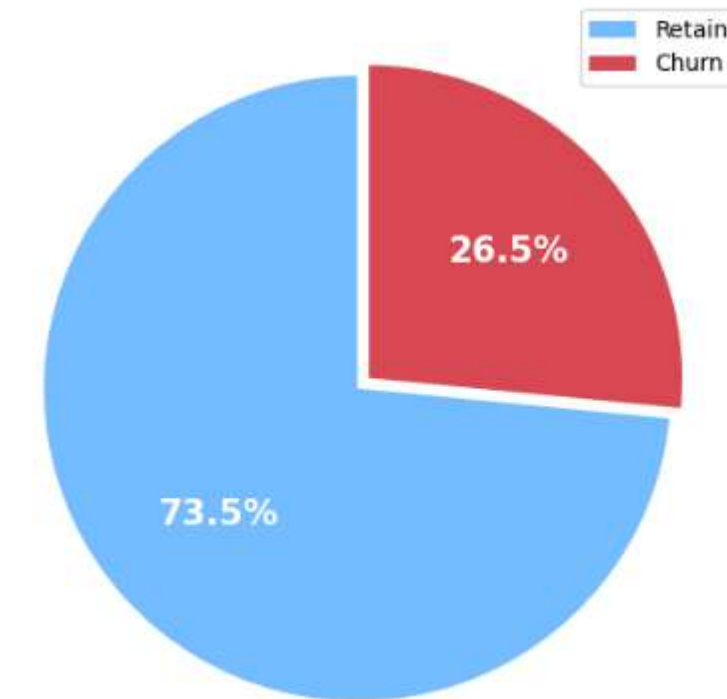
- Customer acquisition costs are 5–25x higher than retention, yet current efforts are reactive rather than proactive.
- Broad discount application is causing unnecessary margin erosion without guaranteed loyalty.
- Lack of customer segmentation means high-value at-risk customers receive the same treatment as low-value ones.
- The company is investing in low-impact retention while high-value assets exit the business undetected.



QUANTIFYING THE LEAKAGE

Churn Impact

- 1,869 customers lost in the current period
- 26.5% churn rate — more than 1 in 4 customers leaving



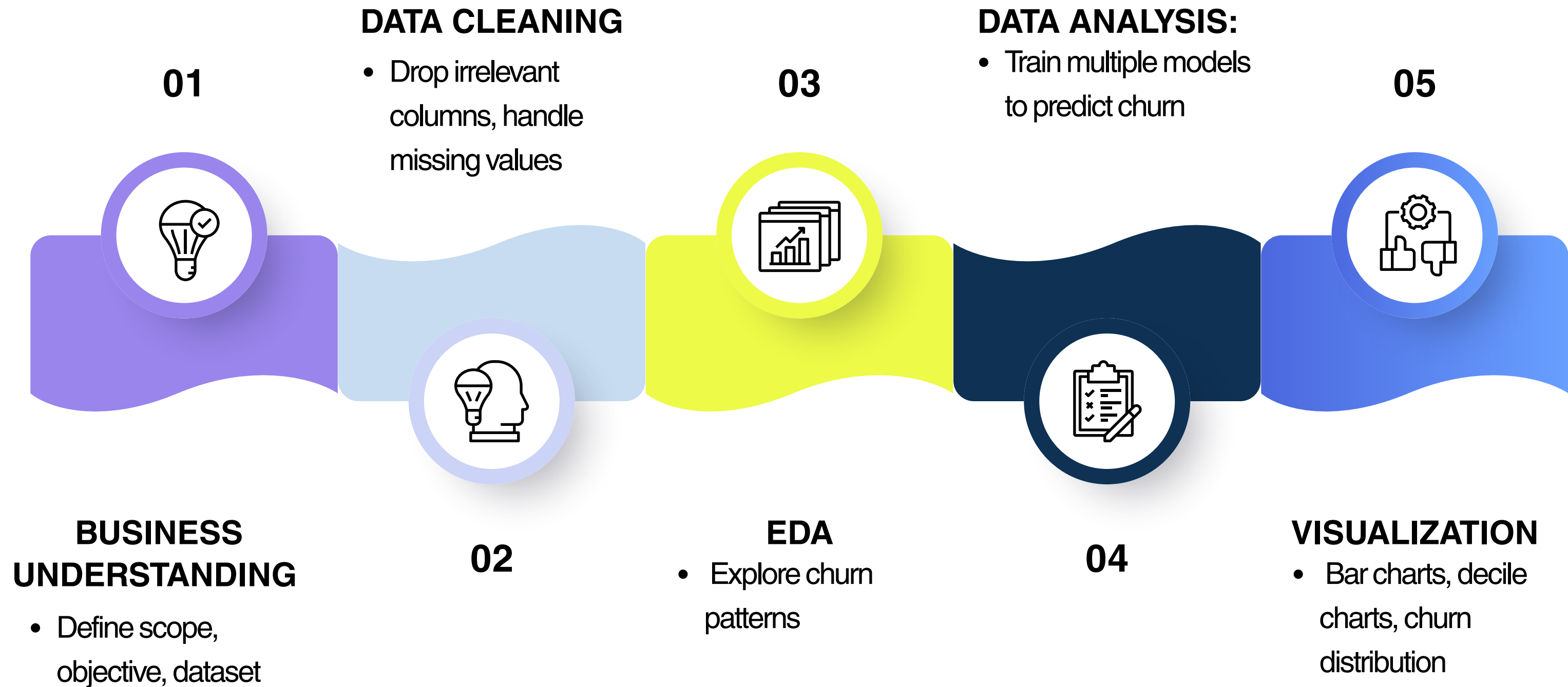
- Churned customers pay 21.5% more per month on average than those who stay. We are bleeding our premium margin.

OBJECTIVES



- Understanding the customer behaviors:
- What makes customers leave the service
- What keeps customers using the service
- Reduce churn for Q4 by predicting high-risk customers and applying retention strategies

METHODOLOGY



KEY DRIVERS OF CHURN

Using SHAP feature importance, three core drivers of churn is identified.



CONTRACT TYPE

43%

Month-to-Month customers
churned



MONTHLY CHARGES

36.%

Customers with high monthly
charges churned



TENURE GROUP

47%

Customers with last than 1 year
tenure churned

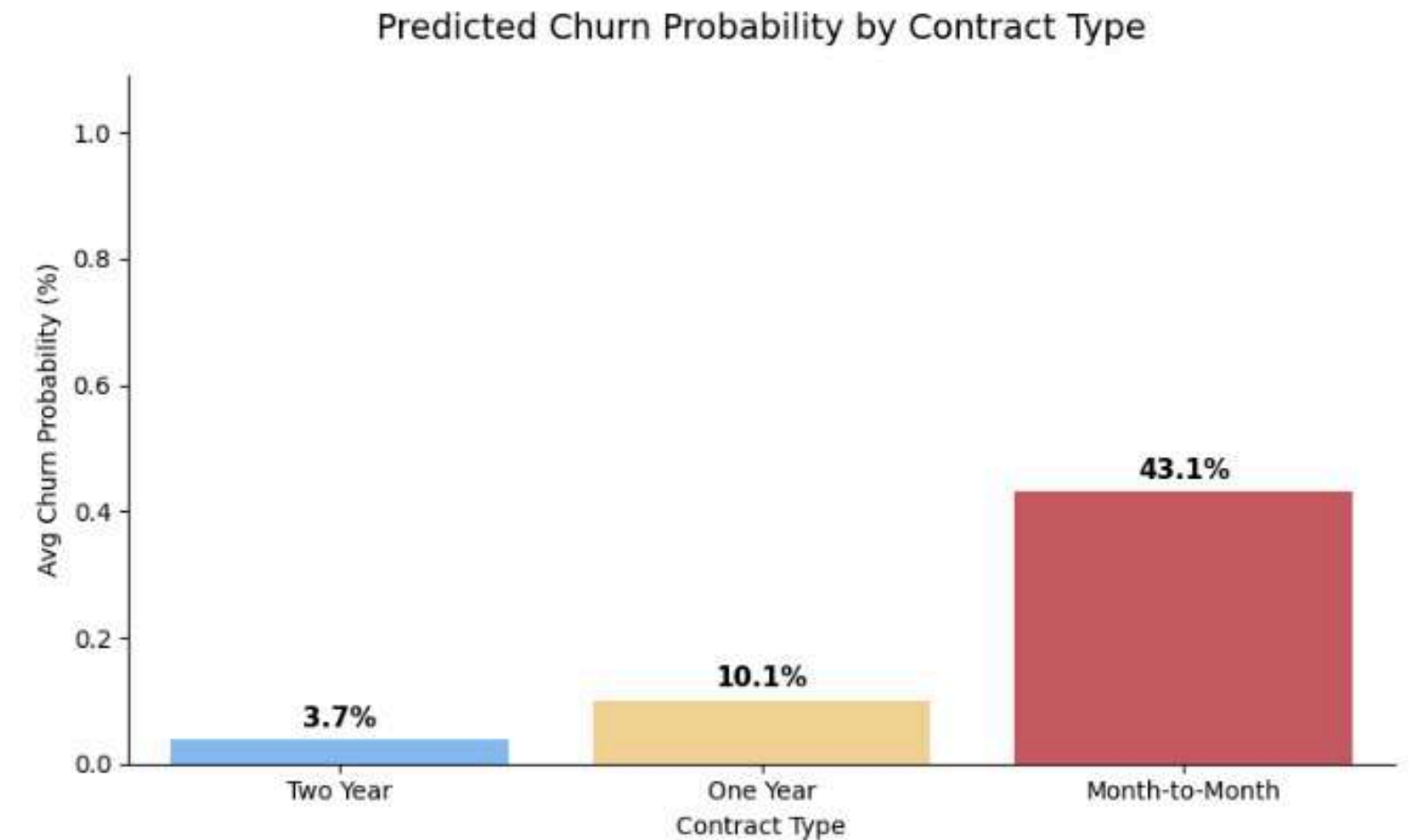
CONTRACT TYPE

Month-to-Month: High Risk (43%) – Extreme volatility.

One-Year: Moderate Risk (10%) – Stable baseline.

Two-Year: Low Risk (3%) – Maximum retention.

Customers on non-committal plans are **3–4x more likely to churn**, highlighting a critical need to incentivize contract migrations.



MONTHLY CHARGES

Churn Risk by Monthly Charges

High Charges(>\$70): 36% Churn (Critical Risk)

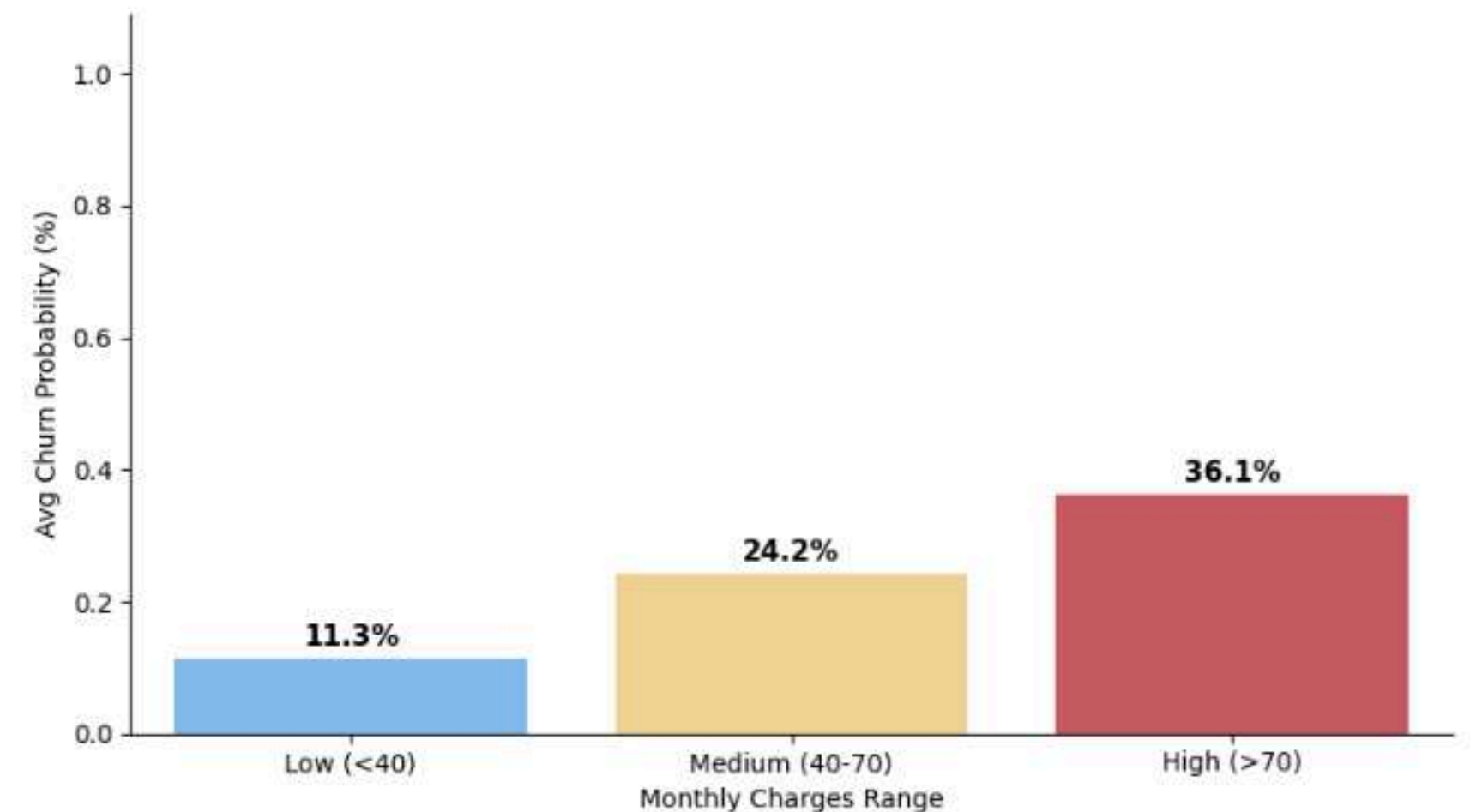
Medium Charges (\$40–\$70): 24% Churn (Moderate Risk)

Low Charges(<\$40): 11% Churn (High Price Value)

Insight:

Churn scales with spend. Higher price points trigger a perceived value gap, making premium customers our most volatile segment.

Predicted Churn by Monthly Charges



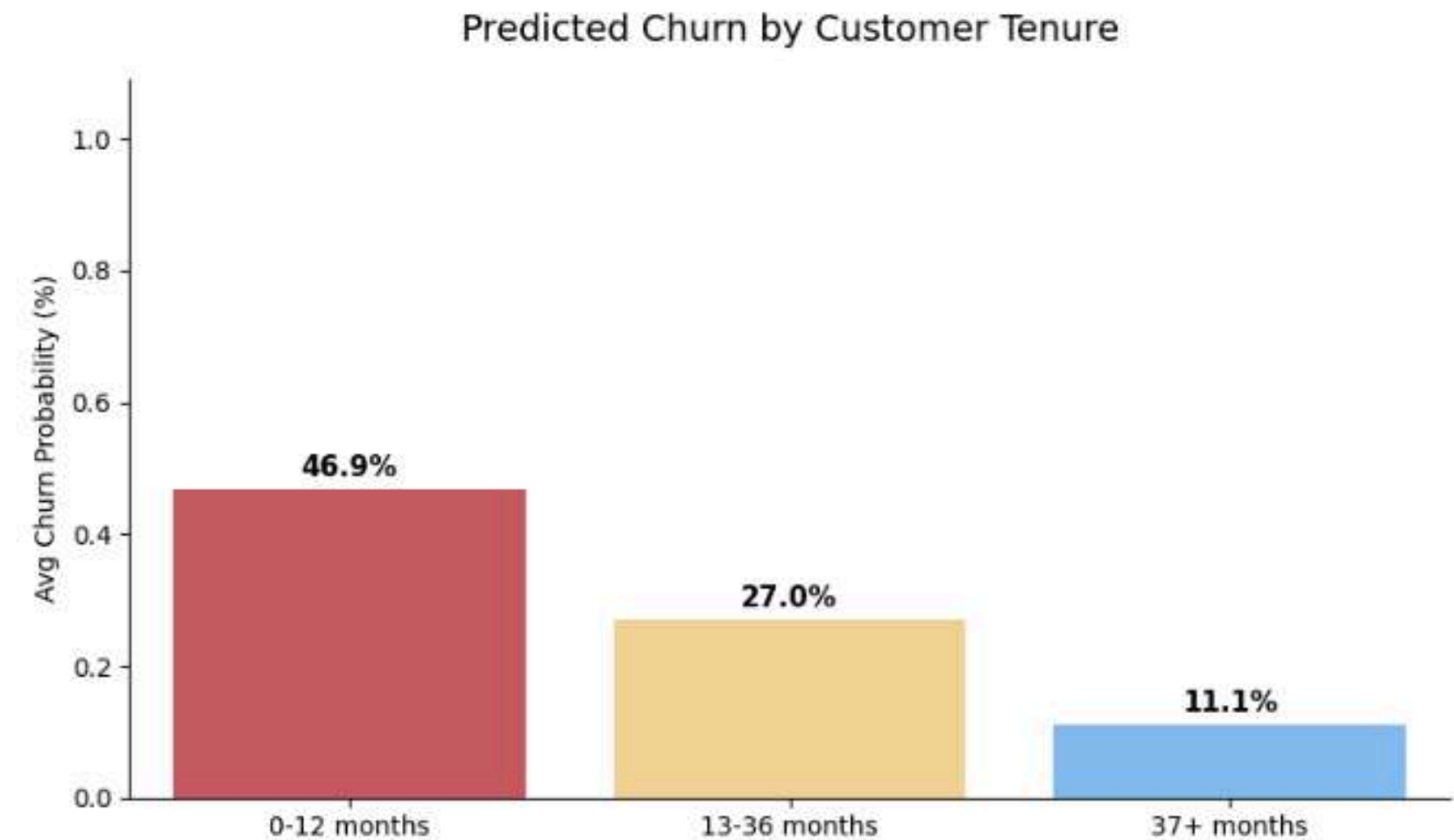
TENURE GROUP

0–12 months : Very high churn (~40–50%)

12–36 months : Medium (~20–30%)

>37 months : Low (<10–15%)

Insight: The **first year** is the **most critical retention window**. Focus on giving proactive technical support/services during this period

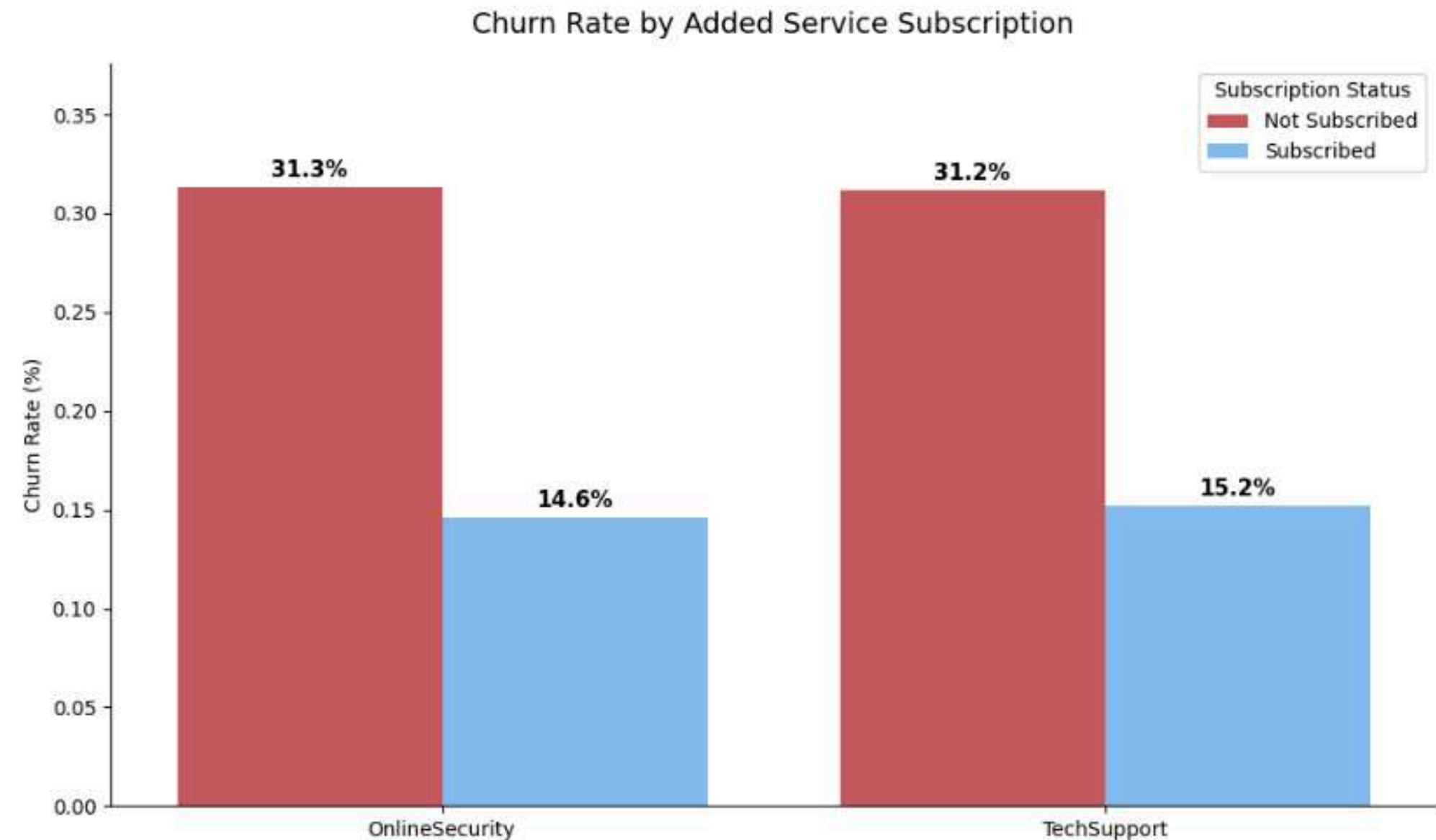


CHURN RETENTION DRIVER

Customers who only subscribe to core internet services are significantly more likely to churn.

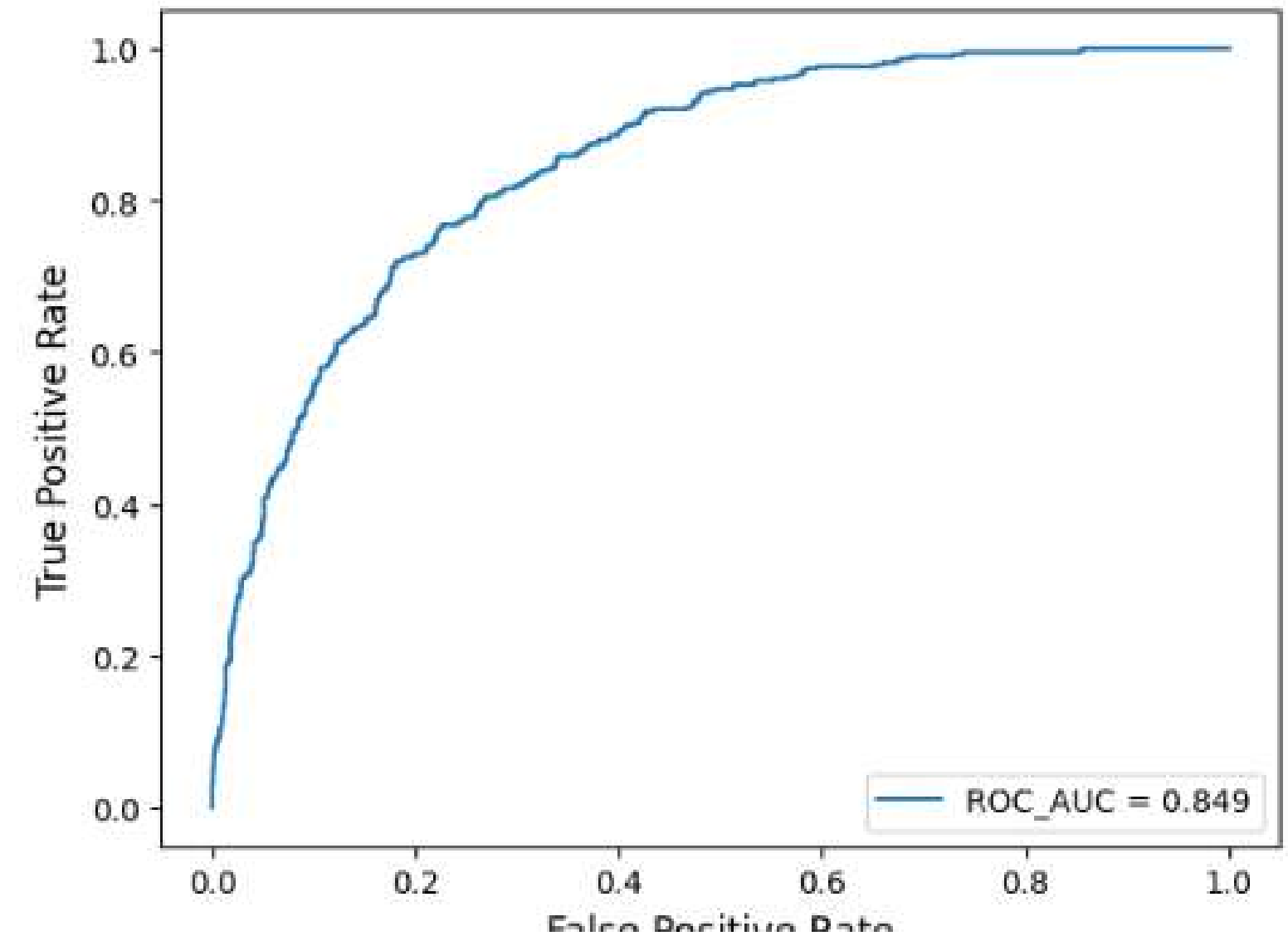
In both categories, subscribing to the service correlates with **cutting the churn rate by more than half**.

Rather than relying on costly cash discounts, introduce targeted service bundles or free trials (e.g., Tech Support, Security).



PREDICTIVE INTELLIGENCE ENGINE

- Model: CatBoost (best performer)
- Accuracy: ~81%
- Recall: up to 82% (after tuning)
- Vast majority of churn identified before it happens.



CUSTOMER SEGMENTS

SEGMENT 1: HIGH VALUE + HIGH RISK

Drivers:

- Month-to-month contract
- High monthly charges
- Short tenure

They are new expensive customers with no lock-in

SEGMENT 2: LOW VALUE + HIGH RISK

Drivers:

- No Tech Support
- No Online Security
- Low engagement

They are unattached users
(easy to leave)

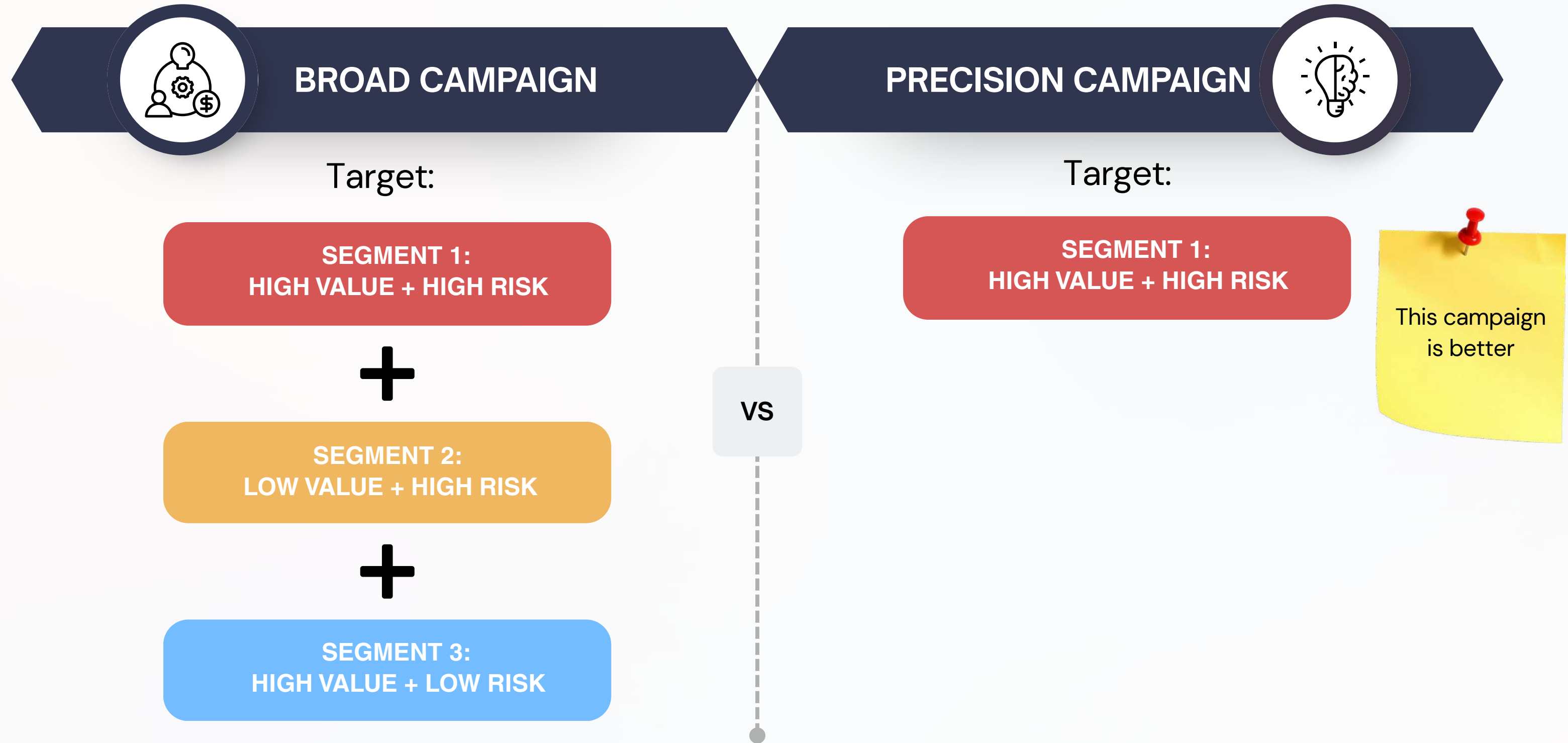
SEGMENT 3: HIGH VALUE + LOW RISK

Drivers:

- Two-year contract
- Longer tenure

They are stable customers,
protect them

STRATEGY APPROACH



WHY PRECISION CAMPAIGN?

Churns are **not equal**

For example:

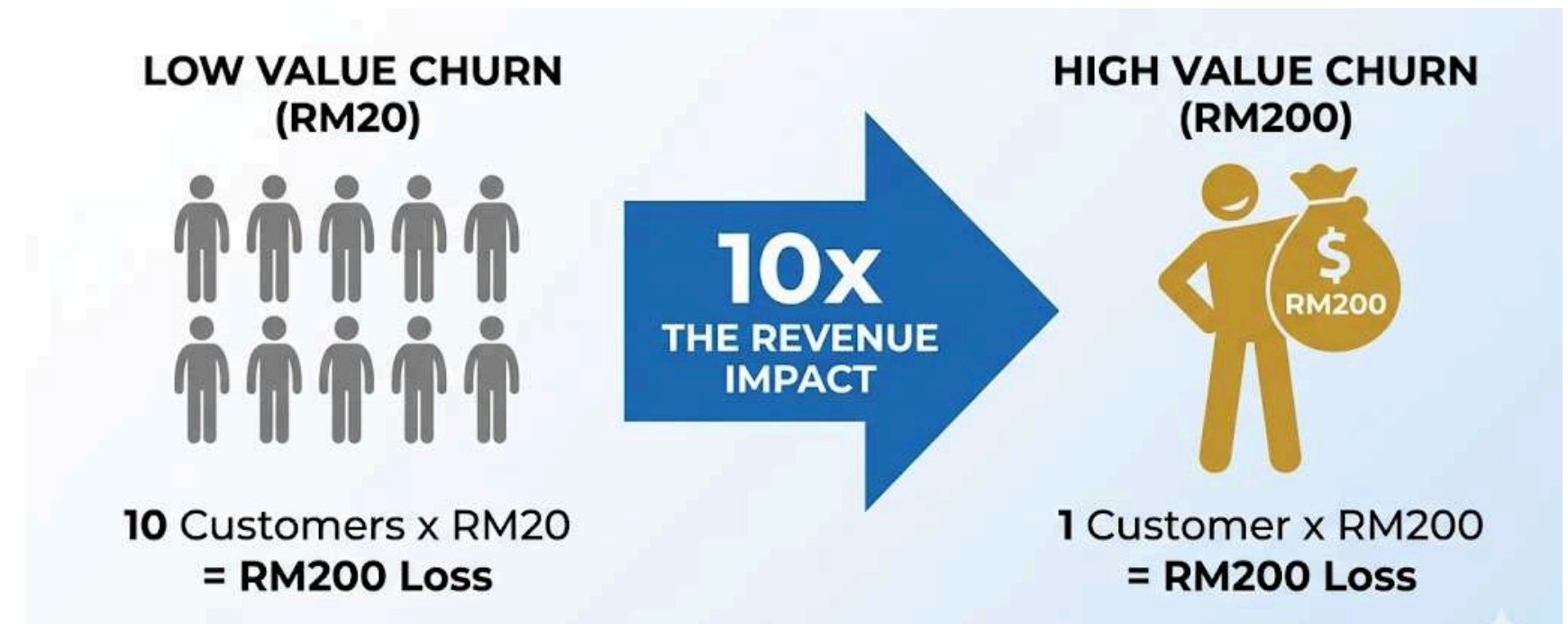
- Losing a RM200 customer is 10 times more damaging than losing a RM20 customer.

Targeted Analysis:

- The strategy focuses on the intersection of Churn Risk (Probability) and Customer Value (Monthly Charges).

Strategic Prioritization :

- Resources are focused on customers who are **both high-risk and high-value**.
- This approach targets the areas of greatest potential revenue loss to protect the bottom line.



CRITICAL REVENUE RISK ZONE

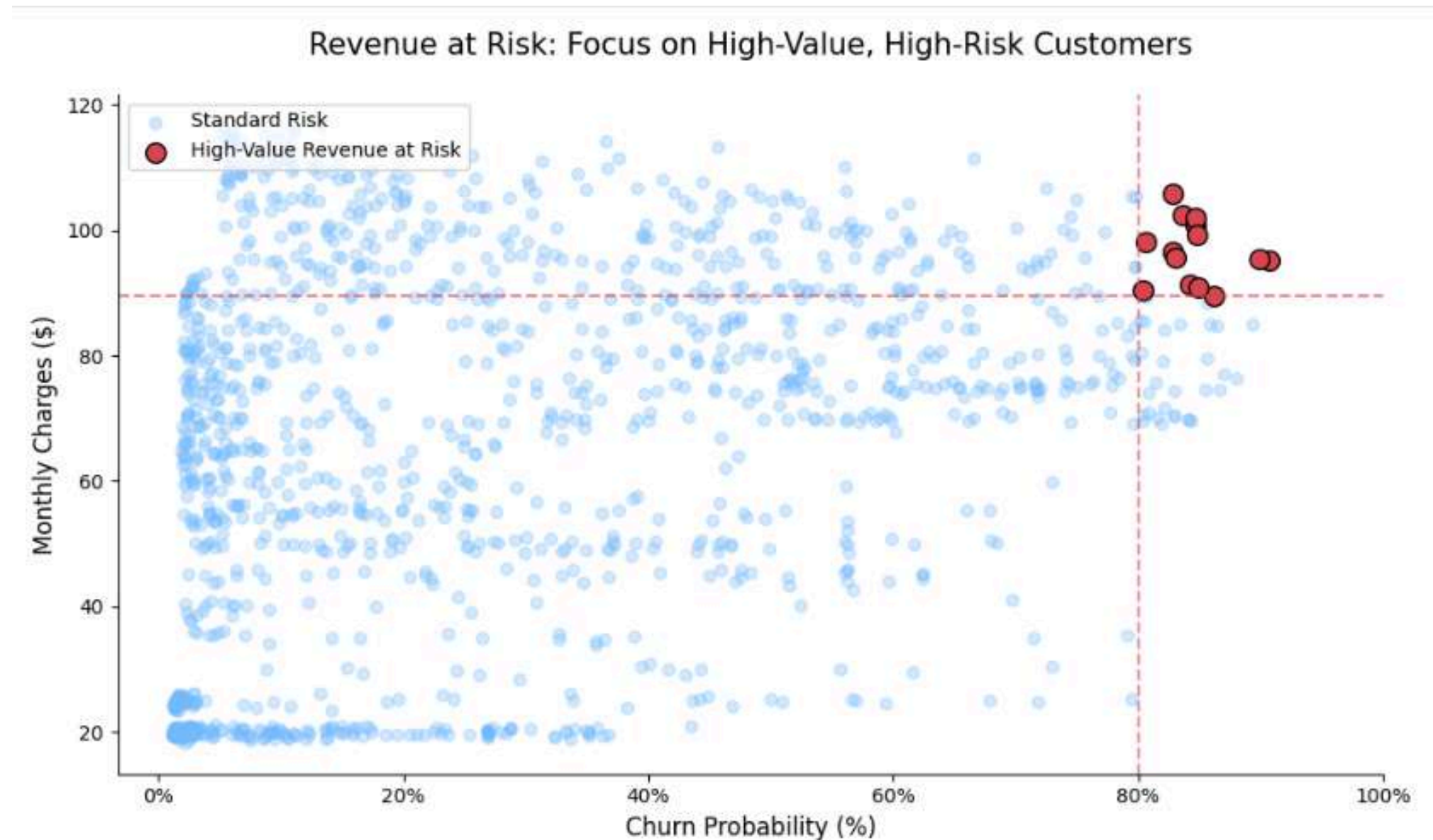
High-Value + High-Risk Customers identified

Definition:

- Churn probability > 80%
- Top 25% monthly revenue (> \$90)

Insight:

- Small group, large revenue impact
- Losing 1 “whale” = 4–5 normal customers



TARGETED RETENTION STRATEGY

SEGMENT 1: HIGH VALUE + HIGH RISK

Goal: Prevent immediate revenue loss

- 10% loyalty discounts
- Offer contract upgrade incentives (month-to-year conversion)
- Bundle premium services (Tech Support + Security trial)



SEGMENT 2: LOW VALUE + HIGH RISK

Goal: Enhance customer retention cost-effectively

- Free Tech Support trial (30 days)
- Digital onboarding education campaign
- Reduce friction in billing/payment issues



SEGMENT 3: HIGH VALUE + LOW RISK

Goal: Protect profitable customers

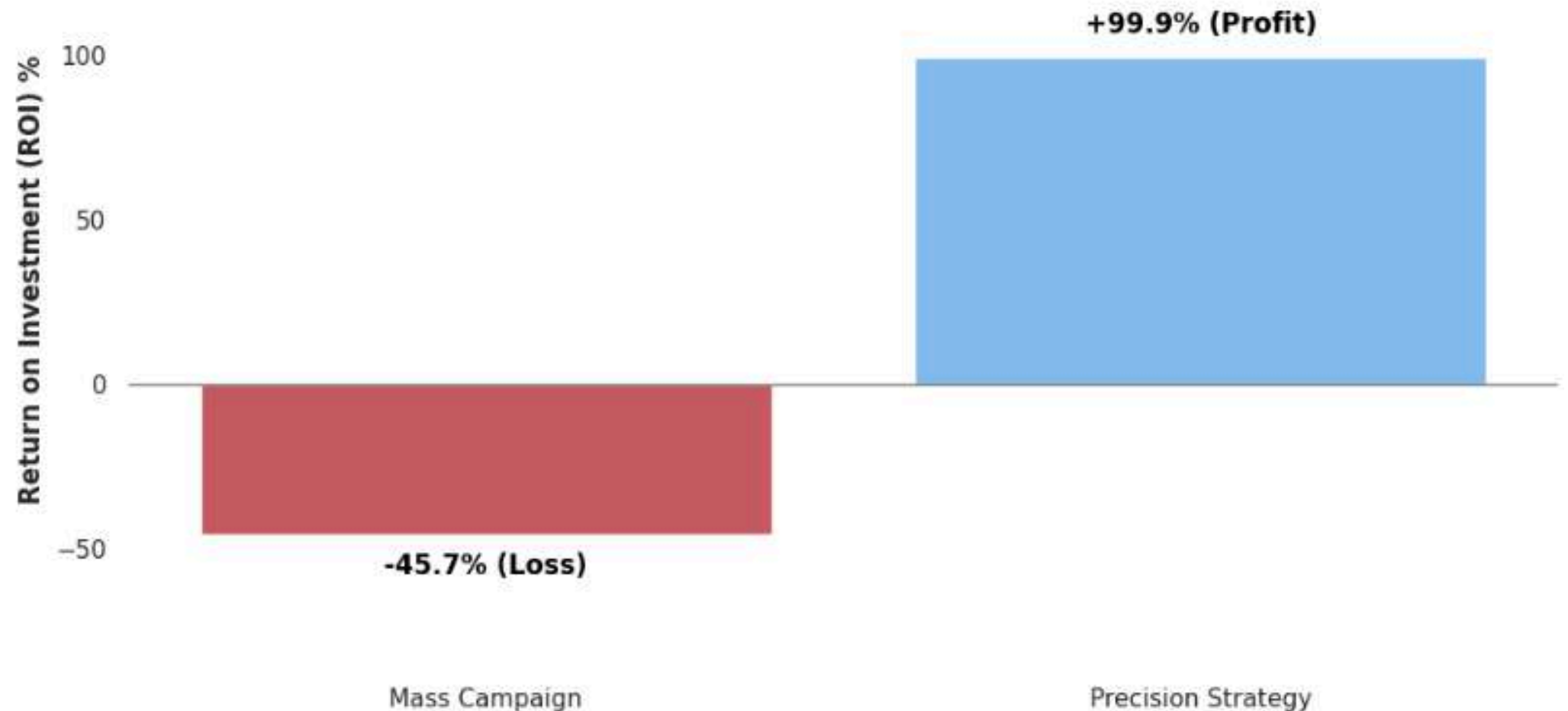
- Loyalty rewards program
- Early renewal discounts
- VIP customer support priority



Note: ROI calculation assumes a 10% targeted loyalty discount with a 25% retention success rate.

ROI SIMULATION

- Mass Discounting Approach:
- 46% Loss on Investment due to "Incentive Waste."
- Result: Revenue is lost by subsidizing customers who would have stayed anyway.
- Precision Retention Approach:
- 100% Profit on Investment by targeting only High-Value/High-Risk "Whales."
- Result: Maximizes profit by spending only where intervention is necessary.



RETENTION STRATEGY COMPARISON

Characteristic	Broad Strategy (Baseline)	Precision Strategy (Optimized)
Focus	Volume (Catch everyone)	Value (Protect the Whales)
Target Count	191 Customers	20 Customers
Incentive Waste	High (Subsidizes stayers)	Minimal (Surgical application)
Threshold	Low Confidence	High Confidence (>80%)
Net Profit	-\$1,405.18 (Loss)	+\$200.40 (Profit)
ROI	-46.00%	100.00%

LINKS

1. [Dataset from Kaggle](#)
2. [Python Code \(Google Colab\)](#).
3. [Power BI Dashboard](#)